Mark Scheme (Results)

Mock Set 1

Pearson Edexcel GCE
In AS Mathematics (8MA0) Paper 22

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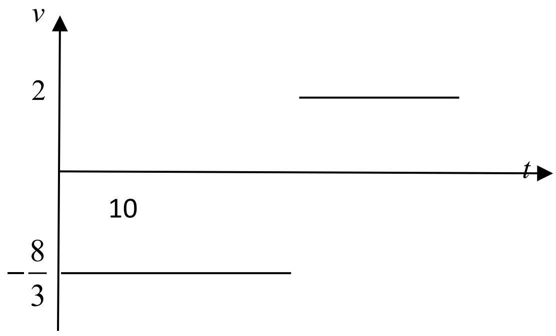
General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Question	Scheme	Marks	AOs
1(a)	Speed = $\frac{8}{3}$ oe (m s^{-1})	B1	1.1b
		(1)	
1(b)	Average speed = $\frac{16+8}{10}$	M1	3.1a
	2.4 (m s^{-1})	A1	1.1b
		(2)	
1(c)		B1 Shape	1.1b
		B1**ft** figs	1.1b
		(2)	
1(d)(i)	The velocity (or speed or direction of motion) (of car) changes	B1	2.4
1(d)(ii)	In reality this cannot occur instantaneously	B1	3.5b
		(2)	
			(7 marks)

Notes:

1a	B1	Accept 2.7 or better. Must be positive
1b	M1	Clear attempt to find $\frac{\text{sum of two distances}}{10}$ (must be positive)
	A1	cao
1c	B1	Two horizontal lines, one above and one below the t-axis. B0 if solid vertical line is included at $t = 6$ and/or 10
	B1**ft**	Graph could be reflected in the t-axis giving, -2 and $\frac{8}{3}$ on the v-axis, ft on their $\frac{8}{3}$
1di	B1	Any equivalent statement
1dii	B1	Any equivalent statement

Question	Scheme	Marks	AOs
2.	$(1-c)\mathbf{i}+(2-10)\mathbf{j}$	M1	1.1b
	$(1-c)\mathbf{i}+(2-10)\mathbf{j}=5\mathbf{a}$	M1	3.1a
	Use of magnitude of resultant = 10 or magnitude of **a** = 2	DM1	2.1
	$(1-c)^2+(2-10)^2=10^2$ oe	A1	1.1b
	$c=7$	A1	1.1b
		(5)	
			(5 marks)

Notes: Allow use of column vectors

2.	M1	Clear attempt to add the forces, seen or implied. Must collect **i**'s and **j**'s
	M1	Use of **F** = 5**a**
	DM1	Must have a quadratic in c only, dependent on first M1
	A1	Correct unsimplified quadratic in c
	A1	A0 if they do not clearly reject $c=-5$

Question	Scheme	Marks	AOs
3(a)	Differentiate v wrt t	M1	3.1a
	$6t^{\frac{1}{2}} - 6t$ (m s^{-2})	A1	1.1b
		(2)	
3(b)	Integrate v wrt t	M1	3.1a
	$6t + \frac{8}{5}t^{\frac{5}{2}} - t^3$ $(+C)$	A1	1.1b
	$6t^{\frac{1}{2}} - 6t = 0$ and attempt to solve	M1	1.1b
	$t = 1$	A1	1.1b
	Substitute their t value into their s expression	M1	1.1b
	$\frac{33}{5}$ oe (m)	A1	1.1b
		(6)	
		(8 marks)	

Notes:

3a	M1	Both powers decreasing by 1
	A1	Correct unsimplified expression
3b	M1	At least two powers increasing by 1
	A1	Correct unsimplified expression
	M1	Equate their acceleration to zero (must have differentiated) and solve for t
	A1	cao
	M1	Must have integrated and equated acceleration to zero
	A1	cao

Question	Scheme	Marks	AOs
4(a)	Equation of motion for A	M1	3.3
	$T - \frac{1}{2}mg = 2ma$	A1	1.1b
		(2)	
4(b)	Equation of motion for B **OR** whole system equation	M1	3.4
	$3mg - T = 3ma$ **OR** $3mg - \frac{1}{2}mg = 5ma$	A1	1.1b
	Solve equations for a	DM1	1.1b
	$a = \frac{1}{2}g$	A1	1.1b
	$V = \sqrt{2 \times \frac{1}{2}g \times h}$ or $V^2 = 2 \times \frac{1}{2}g \times h$	M1	3.1a
	$V = \sqrt{gh}$	A1	1.1b
		(6)	
4(c)	V is greater than W, since air resistance will reduce the size of a	B1	3.5a
		(1)	
4(d)	e.g. include the weight or extensibility of the string	B1	3.5c
		(1)	
			(10 marks)

Notes:

4a	M1	Translate situation into the model and set up the equation of motion for A, with usual rules
	A1	Correct equation
4b	M1	Translate situation into the model and set up the equation of motion for B, with usual rules or use the whole system equation
	A1	Correct equation
	DM1	Solve for a, dependent on previous M mark
	A1	Allow 4.9 ($\mathrm{m\ s^{-2}}$) at this stage
	M1	Complete method to produce an expression for v or v^2 in terms of g and h
	A1	cao
4c	B1	Any equivalent statement
4d	B1	The inclusion of any incorrect statements scores B0

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